

Contents

1 Basic	
1.1 default	
1.2 奇妙的東西	
1.3 random	
2 Binary Search	
2.1 二分搜	
2.2 三分搜	
3 DataStructure	
3.1 DSU(並查集)	
3.2 FenwickTree/BIT	
3.3 二維 BIT	
3.4 Seg/線段樹	
3.5 李超線段樹	
3.6 SegTree Beats/線段樹 Beats	
4 Graph	
4.1 Topo/拓樸	
4.2 bellman/邊長鬆弛	
4.3 ola/歐拉	
4.4 LCA/最近公共祖先	
4.5 Scc/強連通分量	
4.6 Bcc/雙連通分量	
4.7 最大團	
4.8 平均最小團	
4.9 支配點	
5 dynamic programming	
5.1 SOS DP	
5.2 p-median	
6 Computational Geometry	
6.1 default	
6.2 Circle	
6.3 極座標排序	
6.4 ConvexHull/凸包 (傑)	
6.5 ConvexHull/凸包 (宏)	
6.6 PtSegDist	
6.7 兩線相交	
6.8 點是否在多邊形內	
6.9 最遠點對/選轉卡尺	
6.10 最近點對	
7 Flow	
7.1 km/二分圖最大權匹配	
7.2 Dinic	
7.3 hungarian/匈牙利演算法	
7.4 SPFA 最小費流	
7.5 dijk_flow/最小費流	
7.6 差分約束系統	
8 String	
8.1 KMP	
8.2 Z-value	
8.3 馬拉車	
8.4 minRotation/最小字典序旋轉	
8.5 SAIS	
8.6 Trie/字典樹	
8.7 ACautomata/AC 自動機	
9 酷東西可能會用到	
9.1 areaofrect/矩形聯集面積	
10 Math	
10.1 basic	
10.2 快速冪	
10.3 大數運算	
10.4 Prime	
10.5 ExGCD	
10.6 Phi	
10.7 Euler/歐拉函數	
10.8 miller rabin/判斷質數	
10.9 Pollard Rho/質因數分解	
10.10 FFT/傅立葉轉換 (多項式相乘)	
10.11 josephus/約瑟夫問題	
10.12 指數塔	
10.13 BSGS	
10.14 $x^2 \pmod p == a$	
10.15 Theorem	

1 Basic

1.1 default

```

#include <bits/stdc++.h>
using namespace std;
#define int long long
#define idonthavegirlfriend ios::sync_with_stdio(0),cin
.tie(0);
#define pb push_back
#define endl "\n"
#define all(v) (v).begin(),(v).end()
void solve(){
}
signed main(){
    idonthavegirlfriend
    int _=1;
    //cin>>_;
    while(_--){
        solve();
    }
}

```

1.2 奇妙的東西

```

priority_queue<T> //由大到小
priority_queue<T,vector<T>,greater<T>> //由小到大
__builtin_popcountll // 二進位有幾個1(記得這是long long)
__builtin_clzll // 左起第一個1之前0的個數
__builtin_parityll // 1的個數的奇偶性
__builtin_mul_overflow(a,b,&h) // a*b是否溢位,h = a * b
;
__builtin_add_overflow(a,b,&h)

```

1.3 random

```

mt19937 mt(chrono::steady_clock::now().time_since_epoch
().count());
//mt19937_64 mt() -> return randnum
int randint(int l, int r){
    uniform_int_distribution<> dis(l, r); return dis(mt
);
}

```

2 Binary Search

2.1 二分搜

```

int bsearch_1(int l, int r)
{
    while (l < r)
    {
        int mid = l + r >> 1;
        if (check(mid)) r = mid;
        else l = mid + 1;
    }
    return l;
}
// .....0000000000
int bsearch_2(int l, int r)
{
    while (l < r)
    {
        int mid = l + r + 1 >> 1;
        if (check(mid)) l = mid;
        else r = mid - 1;
    }
    return l;
}
// 000000000.....
int m = *ranges::partition_point(views::iota(0LL,(int)1
e9+9),[&](int a){
    return check(a) > k;
});
//[begin,last)
//1111111000000000000
//搜左邊數過來第一個 0
//都是 1 會回傳 last

```

```

int partitionpoint(int L,int R,function<bool(int)> chk)
{
    int l = L,r = R-1;
    while(r - l > 10){
        int m = l + (r-l)/2;
        if(chk(m)) l = m;
        else r = m;
    }
    int m = l;
    while(m <= r){
        if(!chk(m)) break;
        ++m;
    }
    if(!chk(m)) return m;
    else return R;
}
//手工

```

2.2 三分搜

```

int l = 1,r = 100;
while(l < r) {
    int lmid = l + (r - l) / 3; // l + 1/3區間大小
    int rmid = r - (r - l) / 3; // r - 1/3區間大小
    lans = cal(lmid),rans = cal(rmid);
    // 求凹函數極小值
    if(lans <= rans) r = rmid - 1;
    else l = lmid + 1;
}

```

3 DataStructure

3.1 DSU(並查集)

```

struct DSU {
    vector<int> f, sz;
    int cnt;
    DSU(int n) : f(n), sz(n) {
        for (int i = 0; i < n; i++) {
            f[i] = i;
            sz[i] = 1;
        }
    }
    int find(int x) {
        if (x == f[x]) return x;
        return f[x] = find(f[x]);
    }
    void merge(int x, int y) {
        x = find(x); y = find(y);
        if (x == y) return;
        if (sz[x] < sz[y])
            swap(x, y);
        f[y] = x;
        sz[x] += sz[y];
        cnt--;
    }
    bool same(int a, int b) {
        return find(a) == find(b);
    }
    int groups() { //找有幾群
        return cnt;
    }
};

```

3.2 FenwickTree/BIT

```

struct Fenwick { //這個是1base的
    int n;
    vector<int> s;
    int lowbit(int x) { return x & -x; }
    Fenwick(int _n) {
        n = _n + 1;
        s.clear(), s.resize(n, 0);
    }
    void update(int i, int v) { // 更新數值
        for (; i < n; i += lowbit(i))
            s[i] += v;
    }
    int query(int x) { // 查詢前綴和(有包含)
        int pre = 0;
        for (; x; x -= lowbit(x))

```

```

            pre += s[x];
        return pre;
    }
    Fenwick(vector<int> a) { // 建構資料結構
        n = a.size(); // a 是 0-based
        s.clear(), s.resize(n + 1, 0); // s 是 1-based
        for (int i = 1; i <= n; i++)
            update(i, a[i - 1]);
    }
};
//使用:
Fenwick fw(a); //a is vector
Fenwick fw(n); //n is int

```

3.3 二維 BIT

```

struct BIT {
    int n;
    vector<vector<ll>> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) {
        bit.assign(n, vector<ll>(n, 0));
    }
    void update(int x, int y, ll val) { // 更新 (x, y)
        for (int i = x; i < n; i += lowbit(i))
            for (int j = y; j < n; j += lowbit(j))
                bit[i][j] += val;
    }
    ll query(int x, int y) { // 查詢 (1, 1) ~ (x, y) 的和
        ll ans = 0;
        for (int i = x; i; i -= lowbit(i))
            for (int j = y; j; j -= lowbit(j)) ans += bit[i][j];
        return ans;
    }
    ll range_query(int a, int b, int x, int y) {
        return query(x, y) - query(x, b - 1) -
            query(a - 1, y) + query(a - 1, b - 1);
    }
};

```

3.4 Seg/線段樹

```

struct Seg {
#define cl(x) (x << 1)
#define cr(x) (x << 1) + 1
    int _n;
    vector<int> f, tag;
    vector<int> arr;
    //Seg(int n) : _n(n), f(4 * n), tag(4 * n) {}
    Seg(vector<int> a) {
        _n = a.size();
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
        for (int i = 0; i < _n; i++) {
            arr[i] = a[i];
        }

        build(1, 0, _n - 1);
    }
    Seg(int x){
        _n = x;
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
    }

    void build(int id, int l, int r) {
        if (l == r) {
            f[id] = arr[l];
            return;
        }
        int mid = (l + r) >> 1;
        build(cl(id), l, mid);
        build(cr(id), mid + 1, r);
        pull(id, l, r);
    }

    void pull(int id, int l, int r) {

```

```

    int mid = (l + r) >> 1;
    push(cl(id), l, mid);
    push(cr(id), mid + 1, r);
    f[id] = f[cl(id)] + f[cr(id)];
}

void push(int id, int l, int r) {
    if (tag[id]) {
        f[id] += tag[id] * (r - l + 1);
        if (l != r) {
            tag[cl(id)] += tag[id];
            tag[cr(id)] += tag[id];
        }
        tag[id] = 0;
    }
}

void update(int id, int l, int r, int x, int v) {
    if (l == r) {
        f[id] = v;
        return;
    }
    int mid = (l + r) >> 1;
    if (x <= mid) update(cl(id), l, mid, x, v);
    if (mid < x) update(cr(id), mid + 1, r, x, v);
    pull(id, l, r);
}

void update(int id, int l, int r, int ql, int qr,
            int v) {
    push(id, l, r);
    if (ql <= l and qr >= r) {
        tag[id] += v;
        return;
    }
    int mid = (l + r) >> 1;
    if (ql <= mid) {
        update(cl(id), l, mid, ql, qr, v);
    }
    if (qr > mid) {
        update(cr(id), mid + 1, r, ql, qr, v);
    }
    pull(id, l, r);
}

int query(int id, int l, int r, int sl, int sr) {
    push(id, l, r);
    if (sl <= l and sr >= r) {
        return f[id];
    }
    int res = 0;
    int mid = (l + r) >> 1;
    int ll = 0;
    int rr = 0;
    if (sl <= mid) {
        ll = query(cl(id), l, mid, sl, sr);
    }
    if (sr > mid) {
        rr = query(cr(id), mid + 1, r, sl, sr);
    }
    res = ll + rr;
    return res;
}
};

```

3.5 李超線段樹

```

//線段樹套上凸包
//我相信我們賽中想得出來
const int inf = numeric_limits<i64>::max()/2;
struct Line {
    // y = ax + b
    i64 a{0}, b{-inf};
    i64 operator()(i64 x) {
        return a * x + b;
    }
};

struct Seg{
    int l, r;
    Seg *ls{}, *rs{};
    Line f{};
};

```

```

Seg(int l, int r) : l(l), r(r) {}
void add(Line g){
    int m = (l+r)/2;
    if (g(m) > f(m)) swap(g, f);
    if(g.b == -inf || r - l == 1) return;
    if(g.a < f.a){
        if(!ls) ls = new Seg(l,m);
        ls->add(g);
    }else{
        if(!rs) rs = new Seg(m,r);
        rs->add(g);
    }
}

i64 qry(i64 x){
    int m = (l+r) / 2;
    i64 y = f(x);
    if(x < m && ls) y = max({y,ls->qry(x)});
    if(x >= m && rs) y = max({y,rs->qry(x)});
    return y;
}

};

auto add = [&](Line g,int ql,int qr){ //新增線段 [ql,qr)
}
auto find = [&](auto &&self,Seg * now,int l,int r)
-> void {
    if(ql <= l && r <= qr){
        now->add(g);
        return;
    }
    int m = (l+r) / 2;
    if(ql < m) {
        if(!now->ls) now->ls = new Seg(l,m);
        self(self,now->ls,l,m);
    }
    if(qr > m){
        if(!now->rs) now->rs = new Seg(m,r);
        self(self,now->rs,m,r);
    }
}

find(find,st,-ninf,ninf);
};

//Seg *st = new Seg(-ninf,ninf); // [l,r)

```

3.6 SegTree Beats/線段樹 Beats

```

//阿傑不知道這是啥
struct SegTreeBeat{
    struct node{int mx,smx,cntmx,sum;};
    #define cl(x) (x<<1)+1
    #define cr(x) (x<<1)+2
    #define mid ((l+r)>>1)
    vector<node> seg;
    vector<int> tag;
    vector<int> *a;
    int n;
    void pull(int id){
        seg[id].sum=seg[cl(id)].sum+seg[cr(id)].sum;
    }
    if(seg[cl(id)].mx==seg[cr(id)].mx){
        seg[id].mx=seg[cl(id)].mx;
        seg[id].smx=max(seg[cl(id)].smx,seg[cr(id)].smx);
    }
    seg[id].cntmx=seg[cl(id)].cntmx+seg[cr(id)].cntmx;
}
else if(seg[cl(id)].mx>seg[cr(id)].mx){
    seg[id].mx=seg[cl(id)].mx;
    seg[id].smx=max(seg[cl(id)].smx,seg[cr(id)].mx);
    seg[id].cntmx=seg[cl(id)].cntmx;
}
else{
    seg[id].mx=seg[cr(id)].mx;
    seg[id].smx=max(seg[cr(id)].smx,seg[cl(id)].mx);
    seg[id].cntmx=seg[cr(id)].cntmx;
}
}

void init(vector<int> &v){
    n=v.size();
    seg.resize(n*4+5);
    tag=vector<int>(n*4+5,0);
    a=&v;
}

```

```

void build(int l, int r, int id=0){
    if(l==r){
        seg[id]={(*a)[l], LLONG_MIN, 1, (*a)[l]};
        return;
    }
    build(l, mid, cl(id));
    build(mid+1, r, cr(id));
    pull(id);
}
void apply_chmin(int id, int v){
    if(seg[id].mx<=v) return;
    seg[id].sum-= (seg[id].mx-v)*seg[id].cntmx;
    seg[id].mx=v;
}
void apply_add(int id, int l, int r, int v){
    seg[id].sum+=v*(r-l+1);
    if(seg[id].smx!=LLONG_MIN) seg[id].smx+=v;
    seg[id].mx+=v;
    tag[id]+=v;
}
void push(int id, int l, int r){
    if(l==r) return;
    if(tag[id]){
        apply_add(cl(id), l, mid, tag[id]);
        apply_add(cr(id), mid+1, r, tag[id]);
        tag[id]=0;
    }
    apply_chmin(cl(id), seg[id].mx);
    apply_chmin(cr(id), seg[id].mx);
}
void add(int l, int r, int sl, int sr, int v, int id=0){
    if(sl<=l && r<=sr){
        apply_add(id, l, r, v);
        return;
    }
    push(id, l, r);
    if(sl<=mid) add(l, mid, sl, sr, v, cl(id));
    if(mid<sr) add(mid+1, r, sl, sr, v, cr(id));
    pull(id);
}
void chmin(int l, int r, int sl, int sr, int v, int id=0){
    if(seg[id].mx<=v) return;
    if(sl<=l && r<=sr && seg[id].smx<v){
        apply_chmin(id, v);
        return;
    }
    push(id, l, r);
    if(sl<=mid) chmin(l, mid, sl, sr, v, cl(id));
    if(mid<sr) chmin(mid+1, r, sl, sr, v, cr(id));
    pull(id);
}
int qry(int l, int r, int sl, int sr, int id=0){
    if(sl<=l && r<=sr) return seg[id].sum;
    push(id, l, r);
    int ans=0;
    if(sl<=mid) ans+=qry(l, mid, sl, sr, cl(id));
    if(mid<sr) ans+=qry(mid+1, r, sl, sr, cr(id));
    return ans;
}
};

```

4 Graph

4.1 Topo/拓樸

```

vector<int> edge[MAXN], ans;
int deg[MAXN];

void topo(int n) {
    queue<int> q;

    for (int i = 1; i <= n; i++)
        if (!deg[i])
            q.push(i);

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        ans.push_back(u);

        for (int v : edge[u]) {
            deg[v]--;

```

```

        if (!deg[v])
            q.push(v);
    }
}

```

4.2 bellman/邊長鬆弛

```

struct Edge {
    int from, to, weight;
} edge[MAXN];
int dis[MAXN];
void bellman_ford(int n, int m) {
    dis[1] = 0; // 起點的距離一定是 0
    for (int j = 0; j < n - 1; j++) { // n 個節點要跑 n - 1 次
        for (int i = 0; i < m; i++) { // 對於所有邊都嘗試鬆弛
            if (dis[edge[i].to] >
                dis[edge[i].from] + edge[i].weight)
                dis[edge[i].to] =
                    dis[edge[i].from] + edge[i].weight;
        }
    }
}

```

4.3 ola/歐拉

```

//歐拉路徑(無向圖)兩個奇數點或者全偶點
//歐拉路徑(有向圖)條件:起點out=in+1,終點in=out+1,其他點in=out
//歐拉迴路(無向圖)所有點偶數點
//歐拉迴路(有向圖)所有點in=out
vector<int> anse; //邊
vector<int> ansp; //點
vector<bool> vis(m, false);
function<void(int)> dfs=&[(int u){
    while(!mp[u].empty()){
        auto [x,e]=mp[u].back();
        mp[u].pop_back();
        if(vis[e]){
            continue;
        }
        vis[e]=true;
        dfs(x);
        anse.pb(e);
    }
    ansp.pb(u);
}];

```

4.4 LCA/最近公共祖先

```

struct{
    int n, logN;
    vector<vector<int>>> jump;
    vector<int> in, out;
    bool is_ancestor(int x, int y){
        if(x == -1 || y == -1) return 1;
        return in[x] <= in[y] && out[y] <= out[x];
    }
    int query(int x, int y){
        if(is_ancestor(x, y)) return x;
        if(is_ancestor(y, x)) return y;
        for(int i = logN - 1; i >= 0; i--){
            if(!is_ancestor(jump[x][i], y)) x = jump[x][i];
        }
        return jump[x][0];
    }
}
void init(int _n, vector<vector<int>>>&v){ //size, adj array 0base
    n = _n;
    logN = log2(n) + 1;

    jump.resize(n, vector<int>(logN, -1));
    in.resize(n); out.resize(n);

    int timing = 1;
    auto dfs = [&](auto &&self, int x) -> void{
        in[x] = timing++;
        for(auto &i : v[x]){

```

```

        jump[i][0] = x;
        self(self, i);
    }
    out[x] = timing++;
    return;
}; dfs(dfs, 0);

for(int i = 1; i < logN; i++){
    for(int now = 1; now < n; now++){
        if(jump[now][i-1] == -1) continue;
        jump[now][i] = jump[jump[now][i-1]][i-1];
    }
}
}
}LCA;

```

4.5 Scc/強連通分量

//同一團u,v之間有路徑

//bln:在哪個分輪 E原圖

```

struct Scc{
    int n, nScc, vst[MXN], bln[MXN];
    vector<int> E[MXN], rE[MXN], vec;
    void init(int _n){
        n = _n;
        for (int i=0; i<= n; i++){
            E[i].clear(), rE[i].clear();
        }
    }
    void addEdge(int u, int v){
        E[u].PB(v); rE[v].PB(u);
    }
    void DFS(int u){
        vst[u]=1;
        for (auto v : E[u]) if (!vst[v]) DFS(v);
        vec.PB(u);
    }
    void rDFS(int u){
        vst[u] = 1; bln[u] = nScc;
        for (auto v : rE[u]) if (!vst[v]) rDFS(v);
    }
    void solve(){
        nScc = 0;
        vec.clear();
        fill(vst, vst+n+1, 0);
        for (int i=0; i<n; i++){
            if (!vst[i]) DFS(i);
            reverse(vec.begin(), vec.end());
            fill(vst, vst+n+1, 0);
            for (auto v : vec)
                if (!vst[v]){
                    rDFS(v); nScc++;
                }
        }
    }
};

```

4.6 Bcc/雙連通分量

//雙連通分量定義:每個團塊刪除任意一個點都連通,每個割點都屬於多個雙連通分量

//割點出現數量>=2,割邊出現數量=1

```

struct BccVertex {
    int n, nScc, step, dfn[MXN], low[MXN];
    vector<int> E[MXN], sccv[MXN];
    int top, stk[MXN];
    void init(int _n) {
        n = _n;
        nScc = step = 0;
        for (int i = 0; i < n; i++) E[i].clear();
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        E[v].PB(u);
    }
    void DFS(int u, int f) {
        dfn[u] = low[u] = step++;
        stk[top++] = u;
        for (auto v : E[u]) {
            if (v == f) continue;
            if (dfn[v] == -1) {
                DFS(v, u);
                low[u] = min(low[u], low[v]);
            }
        }
    }
};

```

```

        if (low[v] >= dfn[u]) {
            int z;
            sccv[nScc].clear();
            do {
                z = stk[--top];
                sccv[nScc].PB(z);
            } while (z != v);
            sccv[nScc++].PB(u);
        }
    } else
        low[u] = min(low[u], dfn[v]);
}
}
vector<vector<int>> solve() {
    vector<vector<int>> res;
    for (int i = 0; i < n; i++) dfn[i] = low[i] = -1;
    for (int i = 0; i < n; i++)
        if (dfn[i] == -1) {
            top = 0;
            DFS(i, i);
        }
    REP(i, nScc) res.PB(sccv[i]);
    return res;
}
} graph;

```

4.7 最大團

//最大團定義:團內每個點都有一條邊連接到其他點

//O(1.4422^N)

#define N 111

struct MaxClique{ // 0-base

typedef bitset<N> Int;

Int linkto[N], v[N];

int n;

void init(int _n){

n = _n;

for(int i = 0 ; i < n ; i++){

linkto[i].reset(); v[i].reset();

}

void addEdge(int a , int b)

{ v[a][b] = v[b][a] = 1; }

int popcount(const Int& val)

{ return val.count(); }

int lowbit(const Int& val)

{ return val._Find_first(); }

int ans , stk[N];

int id[N] , di[N] , deg[N];

Int cans;

void maxclique(int elem_num, Int candi){

if(elem_num > ans){

ans = elem_num; cans.reset();

for(int i = 0 ; i < elem_num ; i ++)

cans[id[stk[i]]] = 1;

}

int potential = elem_num + popcount(candi);

if(potential <= ans) return;

int pivot = lowbit(candi);

Int smaller_candi = candi & (~linkto[pivot]);

while(smaller_candi.count() && potential > ans){

int next = lowbit(smaller_candi);

candi[next] = !candi[next];

smaller_candi[next] = !smaller_candi[next];

potential --;

if(next == pivot || (smaller_candi & linkto[next]).count()){

stk[elem_num] = next;

maxclique(elem_num + 1, candi & linkto[next]);

}

int solve(){

for(int i = 0 ; i < n ; i++){

id[i] = i; deg[i] = v[i].count();

}

sort(id , id + n , [&](int id1, int id2){

return deg[id1] > deg[id2]; });

for(int i = 0 ; i < n ; i ++)

for(int i = 0 ; i < n ; i ++)

for(int j = 0 ; j < n ; j ++)

if(v[i][j]) linkto[di[i]][di[j]] = 1;

Int cand; cand.reset();

for(int i = 0 ; i < n ; i ++)

cand[i] = 1;

```

    ans = 1;
    cans.reset(); cans[0] = 1;
    maxclique(0, cand);
    return ans;
} } solver;
/*獨立集 與 團 的概念完全相反 最大獨立集去建反圖之後，
求最大團即可得到答案
獨立集定義：獨立集內的任意兩個點之間都 沒有 有邊相 鄰
*/

```

4.8 平均最小團

```

/* minimum mean cycle O(VE) */
//求最小平均權重圈，若無圈則回傳inf
//定義：圈的平均權重 = 圈的總權重 / 圈的邊數
struct MMC{
#define E 101010
#define V 1021
#define inf 1e9
#define eps 1e-6
    struct Edge { int v,u; double c; };
    int n, m, prv[V][V], prve[V][V], vst[V];
    Edge e[E];
    vector<int> edgeID, cycle, rho;
    double d[V][V];
    void init( int _n )
    { n = _n; m = 0; }
    // WARNING: TYPE matters
    void addEdge( int vi , int ui , double ci )
    { e[ m ++ ] = { vi , ui , ci }; }
    void bellman_ford() {
        for(int i=0; i<n; i++) d[0][i]=0;
        for(int i=0; i<n; i++) {
            fill(d[i+1], d[i+1]+n, inf);
            for(int j=0; j<m; j++) {
                int v = e[j].v, u = e[j].u;
                if(d[i][v]<inf && d[i+1][u]>d[i][v]+e[j].c) {
                    d[i+1][u] = d[i][v]+e[j].c;
                    prv[i+1][u] = v;
                    prve[i+1][u] = j;
                }
            }
        }
    }
    double solve(){
        // returns inf if no cycle, mmc otherwise
        double mmc=inf;
        int st = -1;
        bellman_ford();
        for(int i=0; i<n; i++) {
            double avg=-inf;
            for(int k=0; k<n; k++) {
                if(d[n][i]<inf-eps) avg=max(avg,(d[n][i]-d[k][i])/(n-k));
                else avg=max(avg,inf);
            }
            if (avg < mmc) tie(mmc, st) = tie(avg, i);
        }
        fill(vst,0); edgeID.clear(); cycle.clear(); rho.clear();
        for (int i=n; !vst[st]; st=prv[i--][st]) {
            vst[st]++;
            edgeID.PB(prve[i][st]);
            rho.PB(st);
        }
        while (vst[st] != 2) {
            if(rho.empty()) return inf;
            int v = rho.back(); rho.pop_back();
            cycle.PB(v);
            vst[v]++;
        }
        reverse(ALL(edgeID));
        edgeID.resize(SZ(cycle));
        return mmc;
    }
} } mmc;

```

4.9 支配點

```

//支配點
//最後 idom[u] 為點 u 的支配點
struct DominatorTree{ // O(N)
#define REP(i,s,e) for(int i=(s);i<=(e);i++)
#define REPD(i,s,e) for(int i=(s);i>=(e);i--)
    int n , s;

```

```

vector< int > g[ MAXN ] , pred[ MAXN ];
vector< int > cov[ MAXN ];
int dfn[ MAXN ] , nfd[ MAXN ] , ts;
int par[ MAXN ]; //idom[u] s到u的最後一個必經點
int sdom[ MAXN ] , idom[ MAXN ];
int mom[ MAXN ] , mn[ MAXN ];
inline bool cmp( int u , int v )
{ return dfn[ u ] < dfn[ v ]; }
int eval( int u ){
    if( mom[ u ] == u ) return u;
    int res = eval( mom[ u ] );
    if(cmp( sdom[ mn[ mom[ u ] ] ] , sdom[ mn[ u ] ] ))
        mn[ u ] = mn[ mom[ u ] ];
    return mom[ u ] = res;
}
void init( int _n , int _s ){
    ts = 0; n = _n; s = _s;
    REP( i , 1 , n ) g[ i ].clear(), pred[ i ].clear();
}
void addEdge( int u , int v ){
    g[ u ].push_back( v );
    pred[ v ].push_back( u );
}
void dfs( int u ){
    ts++;
    dfn[ u ] = ts;
    nfd[ ts ] = u;
    for( int v : g[ u ] ) if( dfn[ v ] == 0 ){
        par[ v ] = u;
        dfs( v );
    }
}
void build(){
    REP( i , 1 , n ){
        dfn[ i ] = nfd[ i ] = 0;
        cov[ i ].clear();
        mom[ i ] = mn[ i ] = sdom[ i ] = i;
    }
    dfs( s );
    REPD( i , n , 2 ){
        int u = nfd[ i ];
        if( u == 0 ) continue;
        for( int v : pred[ u ] ) if( dfn[ v ] ){
            eval( v );
            if( cmp( sdom[ mn[ v ] ] , sdom[ u ] ) )
                sdom[ u ] = sdom[ mn[ v ] ];
        }
        cov[ sdom[ u ] ].push_back( u );
        mom[ u ] = par[ u ];
        for( int w : cov[ par[ u ] ] ){
            eval( w );
            if( cmp( sdom[ mn[ w ] ] , par[ u ] ) )
                idom[ w ] = mn[ w ];
            else idom[ w ] = par[ u ];
        }
        cov[ par[ u ] ].clear();
    }
    REP( i , 2 , n ){
        int u = nfd[ i ];
        if( u == 0 ) continue;
        if( idom[ u ] != sdom[ u ] )
            idom[ u ] = idom[ idom[ u ] ];
    }
} } domT;

```

5 dynamic programming

5.1 SOS DP

```

// 求子集和 或超集和 -> !(mask & (1 << i))
for(int i = 0; i<(1<<N); ++i) F[i] = A[i]; //預處理 狀態權重

for(int i = 0; i < N; ++i)
for (int s = 0; s < (1<<N) ; ++s)
    if (s & (1 << i))
        F[s] += F[s ^ (1 << i)];

```

//窮舉子集合

```

for(int s = mask; s ; s = (s-1)&mask;)
//放這個做啥我不會用QAQ

```

5.2 p-median

```

void p_Median(){
    for (int i=1; i<=N; ++i)
        for (int j=i; j<=N; ++j){
            m = (i+j)/2, d[i][j] = 0; // m是中位
            // 數，d[i][j]為距離的總和
            for (int k=i; k<=j; ++k) d[i][j] += abs(arr
                [k] - arr[m]);
        }
    for (int p=1; p<=P; ++p)
        for (int n=1; n<=N; ++n){
            dp[p][n] = 1e9;
            for (int k=p; k<=n; ++k)
                if (dp[p-1][k-1] + d[k][n] < dp[p][n]){
                    dp[p][n] = dp[p-1][k-1] + d[k][n];
                    r[p][n] = k; // 從第k個位置往右
                    // 到第j個位置
                }
        }
}

```

6 Computational Geometry

6.1 default

```

int sgn(double x) { return (x > -eps) - (x < eps); }
template<class T>
struct pt{
    T x,y;
    pt(T _x,T _y):x(_x),y(_y){}
    pt():x(0),y(0){}
    pt operator * (T c){ return pt(x * c, y * c); }
    pt operator / (T c){ return pt(x / c, y / c); }
    pt operator + (pt a){ return pt(x + a.x, y + a.y); }
    pt operator - (pt a){ return pt(x - a.x, y - a.y); }
    T operator * (pt a){ return x * a.x + y * a.y; } //
    // 內積
    T operator ^ (pt a){ return x * a.y - y * a.x; } //
    // 外積
    bool operator < (pt a) const { return x < a.x || (x
        == a.x && y < a.y); } // 按照x排
    bool operator==(const pt<T>& a) const { return x ==
        a.x && y == a.y; }
};
template<class T>
ld abs(pt<T> a) { // 返回向量長度
    return sqrt((ld)(a.x * a.x + a.y * a.y)); }
template<class T>
struct Line { // 直線
    pt<T> a, b;
    pt<T> dir() const { return b-a; }
};
template<class T> // 線段
T ori(pt<T> a, pt<T> b, pt<T> c) { return (b - a) ^ (c -
    a); }
template<class T> // 判斷點是否在線段上
bool PtOnSeg(pt<T> p, Line<T> L) {
    return sgn(ori(L.a, L.b, p)) == 0 and sgn((p - L.a)
        * (p - L.b)) <= 0;
}

```

6.2 Circle

```

struct Cir {
    Pt o;
    ld r;
};
bool disjunct(const Cir &a, const Cir &b) { // 分開 ||
    // 外切 -> 1
    return sgn(abs(a.o - b.o) - a.r - b.r) >= 0;
}
bool contain(const Cir &a, const Cir &b) {
    return sgn(a.r - b.r - abs(a.o - b.o)) >= 0;
}

```

6.3 極座標排序

```

// 由+x軸往逆時針排序
struct point {
    int x, y;
};

```

```

long long cross(const point &a, const point &b) {
    return (long long) a.x * b.y - (long long) a.y * b.
        x;
}
bool cmp(const point &a, const point &b) {
    int ah = (a.y < 0 or (a.y == 0 and a.x < 0));
    int bh = (b.y < 0 or (b.y == 0 and b.x < 0));
    if (ah != bh) return ah < bh;
    return cross(a, b) > 0;
}
void argument_sort(vector<point> &points) {
    sort(points.begin(), points.end(), cmp);
}

```

6.4 ConvexHull/凸包 (傑)

```

template<class T>
vector<pt<T>> Hull(vector<pt<T>> P) {
    sort(all(P));
    P.erase(unique(all(P)), P.end());
    P.insert(P.end(), P.rbegin() + 1, P.rend());
    vector<pt<T>> stk;
    for (auto p : P) {
        auto it = stk.rbegin();
        while (stk.rend() - it >= 2 and \
            ori(*next(it), *it, p) <= 0 and \
            (*next(it) < *it) == (*it < p)) {
            it++;
        }
        stk.resize(stk.rend() - it);
        stk.push_back(p);
    }
    stk.pop_back();
    return stk;
}

```

6.5 ConvexHull/凸包 (宏)

```

struct ConvexHull{
    vector<Pt> vec;
    int n;
    void init(vector<Pt> &v){
        vec=v;
        n=v.size();
    }
    int cross(Pt o,Pt a,Pt b){return (a-o)^(b-o);}
    vector<int> point;
    vector<int> area;
    vector<Pt> up,down;
    void build(){
        up.clear();down.clear();
        point=vector<int>(n+1);
        area=vector<int>(n+1);
        int upsum=0;
        int downsum=0;
        for(int i=0;i<n;i++){
            while(up.size()-1>=2 &&
                cross(up[up.size()-1],
                    up[up.size()-2],vec[i])<=0){
                upsum-=cross({0,0},up[up.size()-2],
                    up[up.size()-1]);
                up.pop_back();
            }
            up.push_back(vec[i]);
            if(up.size()-1>1)
                upsum+=cross({0,0},up[up.size()-2],
                    up[up.size()-1]);

            while(down.size()-1>=2 &&
                cross(down[down.size()-1],
                    down[down.size()-2],vec[i])>=0){
                downsum-=cross({0,0},down[down.size()
                    -2],
                    down[down.size()-1]);
                down.pop_back();
            }
            down.push_back(vec[i]);
            if(down.size()-1>1)
                downsum+=cross({0,0},down[down.size()
                    -2],
                    down[down.size()-1]);
        }
    }
}

```

```

        area[i+1]=abs(upsum-downsum);

        if(i+1==1) point[i+1]=1;
        else point[i+1]=up.size()+down.size()-2;
    }
    double getarea(int p){
        return area[p]/2.0;
    }
    int getnum(int p){
        return point[p];
    }
    vector<Pt> getall(){
        vector<Pt> ans;
        for(int i=0;i<down.size();i++)
            ans.push_back(down[i]);
        for(int i=up.size()-2;i>=1;i--)
            ans.push_back(up[i]);
        return ans;
    }
};

```

6.6 PtSegDist

```

template<class T> //計算點到線段的距離
ld PtSegDist(pt<T> p, Line<T> l) {
    ld ans = min(abs(p - l.a), abs(p - l.b));
    if (sgn(abs(l.a - l.b)) == 0) return ans;
    if (sgn((l.a - l.b) * (p - l.b)) < 0) return ans;
    // case1
    if (sgn((l.b - l.a) * (p - l.a)) < 0) return ans;
    // case2
    return min(ans, abs(ori(p, l.a, l.b)) / abs(l.a - l.b)); // case2
}

```

6.7 兩線相交

```

bool isInter(Line l, Line m) {
    if (PtOnSeg(m.a, l) or PtOnSeg(m.b, l) or
        PtOnSeg(l.a, m) or PtOnSeg(l.b, m))
        return true;
    return PtSide(m.a, l) * PtSide(m.b, l) < 0 and
        PtSide(l.a, m) * PtSide(l.b, m) < 0;
}

Pt LineInter(Line l, Line m) { //交點
    double s = ori(m.a, m.b, l.a), t = ori(m.a, m.b, l.b);
    return (l.b * s - l.a * t) / (s - t);
}

```

6.8 點是否在多邊形內

```

template<class T> //判斷點是否在多邊形內
int inPoly(pt<T> p, const vector<pt<T>> &P) {
    const int n = P.size();
    int cnt = 0;
    for (int i = 0; i < n; i++) {
        pt<T> a = P[i], b = P[(i + 1) % n];
        if (PtOnSeg(p, {a, b})) return 1; // on edge
        if ((sgn(a.y - p.y) == 1) ^ (sgn(b.y - p.y) == 1))
            cnt += sgn(ori(a, b, p));
    }
    return cnt == 0 ? 0 : 2; // out, in
}

```

6.9 最遠點對/選轉卡尺

```

double RotatingCalipers(vector<Pt> &v){ //逆時針
    int n=v.size();
    double dist=0;
    if(n==1) return 0;
    if(n==2) return dis(v[0]-v[1]);
    for(int i=0,j=2;i<n;i++){
        Pt a=v[i],b=v[(i+1)%n];
        while(cross(v[j],a,b)<cross(v[(j+1)%n],a,b))
            j=(j+1)%n;
        dist=max(dist,dis(v[j]-a));
        dist=max(dist,dis(v[j]-b));
    }
    return dist;
}

```

6.10 最近點對

```

//最近點對
//O(nlogn)
pair<ld, pair<i32, i32>> ClosestPair(vector<Pt> &P) {
    // ans = dis * dis !!注意ans overflow問題
    if (P.size() == 1) { return {1e200L, {0, 0}}; }
    pair<i32, i32> ansi;
    auto ans = abs2(P[0] - P[1]);
    ansi = {0, 1};
    auto upd = [&](const Pt &a, const Pt &b) {
        auto dis = abs2(a - b);
        if (dis < ans) ans = dis, ansi.FF = a.id, ansi.SS = b.id;
    };
    auto cmpy = [](const Pt &a, const Pt &b) { return a.y < b.y; };

    vector<Pt> t(P.size() + 1);
    auto rec = [&](auto &&self, i32 l, i32 r) {
        if (r - l <= 3) {
            for (i32 i = l; i <= r; i++)
                for (i32 j = i + 1; j <= r; j++) upd(P[i], P[j]);
            sort(P.begin() + l, P.begin() + r + 1, cmpy);
            return;
        }
        i32 m = (l + r) >> 1;
        auto midx = P[m].x;
        self(self, l, m), self(self, m + 1, r);
        i32 tsz = 0;
        inplace_merge(P.begin() + l, P.begin() + m + 1,
            P.begin() + r + 1, cmpy);
        for (i32 i = l; i <= r; i++) {
            if (abs(P[i].x - midx) * abs(P[i].x - midx)
                >= ans) continue;
            for (i32 j = tsz - 1; j >= 0 && (P[i].y - t[j].y) * (P[i].y - t[j].y) < ans; j--)
                upd(P[i], t[j]);
            t[tsz++] = P[i];
        }
    };
    sort(all(P));
    rec(rec, 0, P.size() - 1);
    return make_pair(sqrt(ans), ansi);
}

```

7 Flow

7.1 km/二分圖最大權匹配

```

struct KM { // max weight, for min negate the weights
    int n, mx[MXN], my[MXN], pa[MXN];
    ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
    bool vx[MXN], vy[MXN];
    void init(int _n) { // 1-based, N個節點
        n = _n;
        for(int i=1; i<=n; i++) fill(g[i], g[i]+n+1, 0);
    }
    void addEdge(int x, int y, ll w) { g[x][y] = w; } //左邊的集合節點x連邊右邊集合節點y權重為w
    void augment(int y) {
        for(int x, z; y; y = z)
            x=pa[y], z=mx[x], my[y]=x, mx[x]=y;
    }
    void bfs(int st) {
        for(int i=1; i<=n; ++i) sy[i]=INF, vx[i]=vy[i]=0;
        queue<int> q; q.push(st);
        for(;;) {
            while(q.size()) {
                int x=q.front(); q.pop(); vx[x]=1;
                for(int y=1; y<=n; ++y) if(!vy[y]) {
                    ll t = lx[x]+ly[y]-g[x][y];
                    if(t==0) {
                        pa[y]=x;
                        if(!my[y]){augment(y);return;}
                        vy[y]=1, q.push(my[y]);
                    } else if(sy[y]>t) pa[y]=x, sy[y]=t;
                }
            }
        }
    }
}

```

```

ll cut = INF;
for(int y=1; y<=n; ++y)
    if(!vy[y]&&cut>sy[y]) cut=sy[y];
for(int j=1; j<=n; ++j){
    if(vx[j]) lx[j] -= cut;
    if(vy[j]) ly[j] += cut;
    else sy[j] -= cut;
}
for(int y=1; y<=n; ++y) if(!vy[y]&&sy[y]==0){
    if(!my[y]){augment(y);return;}
    vy[y]=1, q.push(my[y]);
} } }
ll solve(){ // 回傳值為完美匹配下的最大總權重
    fill(mx, mx+n+1, 0); fill(my, my+n+1, 0);
    fill(ly, ly+n+1, 0); fill(lx, lx+n+1, -INF);
    for(int x=1; x<=n; ++x) for(int y=1; y<=n; ++y) //
        1-base
        lx[x] = max(lx[x], g[x][y]);
    for(int x=1; x<=n; ++x) bfs(x);
    ll ans = 0;
    for(int y=1; y<=n; ++y) ans += g[my[y]][y];
    return ans;
} } graph;

```

7.2 Dinic

```

struct Dinic{
    struct Edge{ int v,f,re; };
    int n,s,t,level[MXN];
    vector<Edge> E[MXN];
    void init(int _n, int _s, int _t){
        n = _n; s = _s; t = _t;
        for (int i=0; i<n; i++) E[i].clear();
    }
    void add_edge(int u, int v, int f){
        E[u].PB({v,f,SZ(E[v])});
        E[v].PB({u,0,SZ(E[u])-1});
    }
    bool BFS(){
        for (int i=0; i<n; i++) level[i] = -1;
        queue<int> que;
        que.push(s);
        level[s] = 0;
        while (!que.empty()){
            int u = que.front(); que.pop();
            for (auto it : E[u]){
                if (it.f > 0 && level[it.v] == -1){
                    level[it.v] = level[u]+1;
                    que.push(it.v);
                }
            }
        }
        return level[t] != -1;
    }
    int DFS(int u, int nf){
        if (u == t) return nf;
        int res = 0;
        for (auto &it : E[u]){
            if (it.f > 0 && level[it.v] == level[u]+1){
                int tf = DFS(it.v, min(nf,it.f));
                res += tf; nf -= tf; it.f -= tf;
                E[it.v][it.re].f += tf;
                if (nf == 0) return res;
            }
        }
        if (!res) level[u] = -1;
        return res;
    }
    int flow(int res=0){
        while (BFS())
            res += DFS(s,2147483647);
        return res;
    } } flow;
/*flow.init(n, source, sink); : 初始化點數、源點、匯點
flow.add_edge(u, v, w); : 加入一條點u連到點v容量為w的邊
flow.flow(); : 回傳值為最大流量
*/

```

7.3 hungarian/匈牙利演算法

```

//二分圖最大匹配
//左邊有L個點，右邊有R個點
//使用前請先呼叫add_edge()加入邊
//匈牙利演算法 O(VE)

```

```

//與km演算法不同，匈牙利演算法只會回傳最大匹配數量，無
//法回傳最大權匹配
struct hungarian {
    int L, R;
    vector<bool> vis;
    vector<int> match;
    vector<vector<int>> E;
    hungarian(int l, int r): //左邊有幾個，右邊有幾個
        L(l), R(r),
        vis(l+r+1),
        match(l+r+1,-1),
        E(l+r+1){}
    void add_edge(int l, int r){//左側第幾個(1-base) ,
        //右側第幾個(1-base)
        r = L+r;
        E[l].push_back(r);
        E[r].push_back(l);
    }
    bool dfs(int u){
        for(int i : E[u]){
            if(vis[i]) continue; // 有連通且未拜訪
            vis[i] = true; // 紀錄是否走過
            if(match[i] == -1 || dfs(match[i])){
                match[i] = u; match[u] = i; // 紀錄匹配
                return true;
            }
        }
        return false;
    }
    int solve(){
        int ans = 0;
        for(int i = 1; i <= L; i++){
            fill(vis,0);
            if(dfs(i)) ans++;
        }
        return ans;
    }
};

```

7.4 SPFA 最小費流

```

struct zkwflow{
    static const int maxN=10000;
    struct Edge{ int v,f,re; ll w;};
    int n,s,t,ptr[maxN]; bool vis[maxN]; ll dis[maxN];
    vector<Edge> E[maxN];
    void init(int _n,int _s,int _t){
        n=_n,s=_s,t=_t;
        for(int i=0;i<n;i++) E[i].clear();
    }
    void addEdge(int u,int v,int f,ll w){
        E[u].push_back({v,f,(int)E[v].size(),w});
        E[v].push_back({u,0,(int)E[u].size()-1,-w});
    }
    bool SPFA(){
        fill_n(dis,n,LLONG_MAX); fill_n(vis,n,false);
        queue<int> q; q.push(s); dis[s]=0;
        while (!q.empty()){
            int u=q.front(); q.pop(); vis[u]=false;
            for(auto &it:E[u]){
                if(it.f>0&&dis[it.v]>dis[u]+it.w){
                    dis[it.v]=dis[u]+it.w;
                    if(!vis[it.v]){
                        vis[it.v]=true; q.push(it.v);
                    }
                }
            }
        }
        return dis[t]!=LLONG_MAX;
    }
    int DFS(int u,int nf){
        if(u==t) return nf;
        int res=0; vis[u]=true;
        for(int &i=ptr[u];i<(int)E[u].size();i++){
            auto &it=E[u][i];
            if(it.f>0&&dis[it.v]==dis[u]+it.w&&!vis[it.v]){
                int tf=DFS(it.v,min(nf,it.f));
                res+=tf,nf-=tf,it.f-=tf;
                E[it.v][it.re].f+=tf;
                if(nf==0){ vis[u]=false; break; }
            }
        }
        return res;
    }
};

```

```

pair<int,ll> flow(){
    int flow=0; ll cost=0;
    while (SPFA()){
        fill_n(ptr,n,0);
        int f=DFS(s,INT_MAX);
        flow+=f; cost+=dis[t]*f;
    }
    return{ flow,cost };
} // reset: do nothing
} flow;

```

7.5 dijk_flow/最小費流

//最小費用最大流不能有負權邊

```

struct MinCostMaxFlow{
    typedef int Tcost;
    static const int MAXV = 20010;
    static const int INFf = 1000000;
    static const Tcost INFc = 1e9;
    struct Edge{
        int v, cap;
        Tcost w;
        int rev;
        Edge(){
            Edge(int t2, int t3, Tcost t4, int t5)
                : v(t2), cap(t3), w(t4), rev(t5) {}
    };
    int V, s, t;
    vector<Edge> g[MAXV];
    void init(int n, int _s, int _t){
        V = n; s = _s; t = _t;
        for(int i = 0; i <= V; i++) g[i].clear();
    }
    void addEdge(int a, int b, int cap, Tcost w){
        g[a].push_back(Edge(b, cap, w, (int)g[b].size()));
        g[b].push_back(Edge(a, 0, -w, (int)g[a].size()-1));
    }
    Tcost d[MAXV];
    int id[MAXV], mom[MAXV];
    bool inqu[MAXV];
    queue<int> q;
    pair<int,Tcost> solve(){
        int mxf = 0; Tcost mnc = 0;
        while(1){
            fill(d, d+1+V, INFc);
            fill(inqu, inqu+1+V, 0);
            fill(mom, mom+1+V, -1);
            mom[s] = s;
            d[s] = 0;
            q.push(s); inqu[s] = 1;
            while(q.size()){
                int u = q.front(); q.pop();
                inqu[u] = 0;
                for(int i = 0; i < (int) g[u].size(); i++){
                    Edge &e = g[u][i];
                    int v = e.v;
                    if(e.cap > 0 && d[v] > d[u]+e.w){
                        d[v] = d[u]+e.w;
                        mom[v] = u;
                        id[v] = i;
                        if(!inqu[v]) q.push(v), inqu[v] = 1;
                    }
                }
                if(mom[t] == -1) break ;
                int df = INFf;
                for(int u = t; u != s; u = mom[u])
                    df = min(df, g[mom[u]][id[u]].cap);
                for(int u = t; u != s; u = mom[u]){
                    Edge &e = g[mom[u]][id[u]];
                    e.cap -= df;
                    g[e.v][e.rev].cap += df;
                }
                mxf += df;
                mnc += df*d[t];
            }
            return {mxf,mnc};
        }
    } flow;
}

```

7.6 差分約束系統

- 2-SAT :
 $(a_i \vee a_j) = \text{true} \quad \forall (i, j)$
 對於任意限制 $(x \vee y)$
 建兩條有向邊 (要多編號 $\neg x$)

$x \rightarrow \neg y$ and $y \rightarrow \neg x$
 跑 scc
 $\text{scc.bl}[x] < \text{scc.bl}[\neg x] \Leftrightarrow x \text{ is true}$
 $\text{scc.bl}[\neg x] < \text{scc.bl}[x] \Leftrightarrow x \text{ is false}$
 $\exists x \text{ which } \text{scc.bl}[x] == \text{scc.bl}[\neg x] \Leftrightarrow \text{無解}$

- 差分約束:
 n 個變數及 m 個約束條件
 求滿足所有 $x_j - x_i \leq b_k \quad (i, j \in [1, n], k \in [1, m])$
 的一組 $x_1 \dots x_n$
 可轉成 $x_j - x_i \leq b_k \rightarrow x_j \leq x_i + b_k$
 結論就是使得所有 x_j 變小以滿足上式
 建邊跑 SPFA/Bellman
 要多建起點 s 連到所有 i 且邊權 0, $\text{dis}[s] = 0$
 有負環則無解, 否則起點到所有 i 的距離為一組解
 $x_j - x_i \leq k \Rightarrow \text{addEdge } i \xrightarrow{k} j$
 $x_j - x_i \geq k \Rightarrow \text{addEdge } j \xrightarrow{-k} i$
 $x_j = x_i \Rightarrow \text{addEdge } i \xrightarrow{0} j \text{ and } j \xrightarrow{0} i$

8 String

8.1 KMP

```

int failure[MXN];
vector<int> KMP(vector<int>& t, vector<int>& p){
    vector<int> ret;
    if (p.size() > t.size()) return {};
    for (int i=1, j=failure[0]=-1; i<p.size(); ++i){
        while (j >= 0 && p[j+1] != p[i])
            j = failure[j];
        if (p[j+1] == p[i]) j++;
        failure[i] = j;
    }
    for (int i=0, j=-1; i<t.size(); ++i){
        while (j >= 0 && p[j+1] != t[i])
            j = failure[j];
        if (p[j+1] == t[i]) j++;
        if (j == p.size()-1){
            ret.push_back(i - p.size() + 1);
            j = failure[j];
        }
    }
    return ret;
}
/*
vector<int> prefunc(const string& s){
    int n = s.size();
    vector<int> pi(n);
    for(int i=1, j=0; i<n; ++i){
        j = pi[i-1];
        while(j && s[j] != s[i]) j = pi[j-1]; //往下找藍框
        if(s[j] == s[i]) ++j; //相等會多一格
        pi[i] = j;
    }
    return pi;
}
*/

```

8.2 Z-value

```

int z[MXN];
void Z_value(const string& s) {
    //z[i] = lcp(s[0...],s[i...])
    int i, j, left, right, len = s.size();
    left=right=0; z[0]=len;
    for(i=1; i<len; i++) {
        j=max(min(z[i-left],right-i),0LL);
        for(; i+j<len&&s[i+j]==s[j]; j++);
        z[i]=j;
        if(i+z[i]>right) {
            right=i+z[i];
            left=i;
        }
    }
}

```

8.3 馬拉車

```

//記得s.resize(2*n+1)
//z_value_pal(&s[0],n,a.data());
void z_value_pal(char *s,int len,int *z){
    len=(len<<1)+1;
    for(int i=len-1; i>=0; i--){
        s[i]=i&1?s[i>>1]:'@';
        z[0]=1;
    }
}

```

```

for(int i=1,l=0,r=0;i<len;i++){
    z[i]=i<r?min(z[l+l-i],r-i):1;
    while(i-z[i]>=0&& i+z[i]<len&& s[i-z[i]]==s[i+z[i]])
        ++z[i];
    if(i+z[i]>r) l=i,r=i+z[i];
} }

```

8.4 minRotation/最小字典序旋轉

```

//rotate(begin(s),begin(s)+minRotation(s),end(s))
int minRotation(string s) {
    int a = 0, N = s.size(); s += s;
    rep(b,0,N) rep(k,0,N) {
        if(a+k == b || s[a+k] < s[b+k])
            {b += max(0LL, k-1LL); break;}
        if(s[a+k] > s[b+k]) {a = b; break;}
    } return a;
}

```

8.5 SAIS

```

// 此為後綴陣列模板
// 用法:
// 把要處理的字串轉成整數陣列傳入suffix_array(), 會得到
// SA與H維結果
const int N = 500010; // 有可能要在這就寫原本字串的兩
// 倍長
struct SA{
#define REP(i,n) for ( int i=0; i<int(n); i++ )
#define REP1(i,a,b) for ( int i=(a); i<=int(b); i++ )
    bool _t[N*2];
    int _s[N*2], _sa[N*2], _c[N*2], x[N], _p[N], _q[N*2],
        hei[N], r[N];
    int operator [] (int i){ return _sa[i]; }
    void build(int *s, int n, int m){
        memcpy(_s, s, sizeof(int) * n);
        sais(_s, _sa, _p, _q, _t, _c, n, m);
        mkhei(n);
    }
    void mkhei(int n){
        REP(i,n) r[_sa[i]] = i;
        hei[0] = 0;
        REP(i,n) if(r[i]) {
            int ans = i>0 ? max(hei[r[i-1]] - 1, 0) : 0;
            while(_s[i+ans] == _s[_sa[r[i]-1]+ans]) ans++;
            hei[r[i]] = ans;
        }
    }
    void sais(int *s, int *sa, int *p, int *q, bool *t,
        int *c, int n, int z){
        bool uniq = t[n-1] = true, neq;
        int nn = 0, nmzx = -1, *nsa = sa + n, *ns = s + n,
            lst = -1;
#define MS0(x,n) memset((x),0,n*sizeof(*(x)))
#define MAGIC(XD) MS0(sa, n); \
        memcpy(x, c, sizeof(int) * z); \
        XD; \
        memcpy(x + 1, c, sizeof(int) * (z - 1)); \
        REP(i,n) if(sa[i] && !t[sa[i]-1]) sa[x[s[sa[i]
            ]-1]]++ = sa[i]-1; \
        memcpy(x, c, sizeof(int) * z); \
        for(int i = n - 1; i >= 0; i--) if(sa[i] && t[sa[i]
            ]-1]) sa[--x[s[sa[i]-1]]] = sa[i]-1;
        MS0(c, z);
        REP(i,n) uniq &= ++c[s[i]] < 2;
        REP(i,z-1) c[i+1] += c[i];
        if (uniq) { REP(i,n) sa[--c[s[i]]] = i; return; }
        for(int i = n - 2; i >= 0; i--) t[i] = (s[i]==s[i
            +1] ? t[i+1] : s[i]<s[i+1]);
        MAGIC(REP1(i,1,n-1) if(t[i] && !t[i-1]) sa[--x[s[i]
            ]]=p[q[i]=nn++]=i);
        REP(i, n) if (sa[i] && t[sa[i]] && !t[sa[i]-1]) {
            neq=lst<0||memcmp(s+sa[i],s+lst,(p[q[sa[i]]+1]-sa
                [i])*sizeof(int));
            ns[q[lst=sa[i]]]=nmzx+=neq;
        }
        sais(ns, nsa, p + nn, q + n, t + n, c + z, nn, nmzx
            + 1);
        MAGIC(for(int i = nn - 1; i >= 0; i--) sa[--x[s[p[
            nsa[i]]]] = p[nsa[i]]);
    }
}

```

```

}sa;
int H[ N ], SA[ N ]; // SA: 後綴陣列的length
// H[i]: SA[i]與SA[i -1]的共同子字
// 串(LCP)
void suffix_array(int* ip, int len) {
    // should padding a zero in the back
    // ip is int array, len is array length
    // ip[0..n-1] != 0, and ip[len] = 0
    ip[len++] = 0;
    sa.build(ip, len, 128);
    for (int i = 0; i < len; i++) {
        H[i] = sa.hei[i + 1];
        SA[i] = sa._sa[i + 1];
    }
    // resulting height, sa array \in [0,len)
}

```

8.6 Trie/字典樹

```

struct trie{
    trie *nxt[26];
    int cnt; //紀錄有多少個字串以此節點結尾
    int sz; //有多少字串的前綴包括此節點
    trie():cnt(0),sz(0){
        memset(nxt,0,sizeof(nxt));
    }
};

//創建新的字典樹
trie *root = new trie();
// 0(!s!)
void insert(string& s){
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        now->sz++;
        if(now->nxt[i-'a'] == NULL){
            now->nxt[i-'a'] = new trie();
        }
        now = now->nxt[i-'a']; //走到下一個字母
    }
    now->cnt++;
    now->sz++;
}

// 0(!s!)
int query_prefix(string& s){ //查詢有多少前綴為 s
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        if(now->nxt[i-'a'] == NULL){
            return 0;
        }
        now = now->nxt[i-'a'];
    }
    return now->sz;
}

int query_count(string& s){ //查詢字串 s 出現次數
    trie *now = root; // 每次從根結點出發
    for(auto i:s){
        if(now->nxt[i-'a'] == NULL){
            return 0;
        }
        now = now->nxt[i-'a'];
    }
    return now->cnt;
}

```

8.7 ACautomata/AC 自動機

```

//makefail before query
//用於多模式字串匹配
struct ACautomata{
    struct Node{
        int cnt,i;
        Node *go[26], *fail, *dic;
        Node (){
            cnt = 0; fail = 0; dic = 0; i = 0;
            memset(go,0,sizeof(go));
        }
    }pool[1048576],*root;
    int ans[1048576];
    int nMem,n_pattern;
    Node* new_Node(){

```

```

    pool[nMem] = Node();
    return &pool[nMem++];
}
void init() {
    nMem=0;root=new_Node();n_pattern=0;
    add("");
}
void add(const string &str) { insert(root,str,0); }
void insert(Node *cur, const string &str, int pos){
    for(int i=pos;i<str.size();i++){
        if(!cur->go[str[i]-'a'])
            cur->go[str[i]-'a'] = new_Node();
        cur=cur->go[str[i]-'a'];
    }
    cur->cnt++;
    cur->i=n_pattern++;
}
void make_fail(){
    queue<Node*> que;
    que.push(root);
    while (!que.empty()){
        Node* fr=que.front(); que.pop();
        for (int i=0; i<26; i++){
            if (fr->go[i]){
                Node *ptr = fr->fail;
                while (ptr && !ptr->go[i]) ptr = ptr->fail;
                fr->go[i]->fail=ptr=(ptr?ptr->go[i]:root);
                fr->go[i]->dic=(ptr->cnt?ptr:ptr->dic);
                que.push(fr->go[i]);
            }
        }
    }
}
void query(string s){
    Node *cur=root;
    for(int i=0;i<(int)s.size();i++){
        while(cur&&!cur->go[s[i]-'a']) cur=cur->fail;
        cur=(cur?cur->go[s[i]-'a']:root);
        if(cur->i>=0) ans[cur->i]++;
        for(Node *tmp=cur->dic;tmp;tmp=tmp->dic)
            ans[tmp->i]++;
    }
}
// ans[i] : number of occurrence of pattern i
}AC;

```

9 酷東西可能會用到

9.1 areaofrect/矩形聯集面積

```

struct AreaofRectangles{
#define cl(x) (x<<1)
#define cr(x) (x<<1|1)
    int n, id, sid;
    pair<int,int> tree[MXN<<3]; // count, area
    vector<int> ind;
    tuple<int,int,int,int> scan[MXN<<1];
    void puint(int i, int l, int r){
        if(tree[i].first) tree[i].second = ind[r+1] -
            ind[l]; // 當前區間有被做加值
        else if(l != r){ // 沒有就直接拿兩個子節點的第二
            個數字
            int mid = (l+r)>>1;
            tree[i].second = tree[cl(i)].second + tree[
                cr(i)].second;
        }
        else tree[i].second = 0;
    }
    void upd(int i, int l, int r, int ql, int qr, int v
    ){
        if(ql <= l && r <= qr){
            tree[i].first += v;
            puint(i, l, r); return;
        }
        int mid = (l+r) >> 1;
        if(ql <= mid) upd(cl(i), l, mid, ql, qr, v);
        if(qr > mid) upd(cr(i), mid+1, r, ql, qr, v);
        puint(i, l, r);
    }
    void init(int _n){
        n = _n; id = sid = 0;
        ind.clear(); ind.resize(n<<1);
        fill(tree, tree+(n<<2), make_pair(0, 0));
    }
    void addRectangle(int lx, int ly, int rx, int ry){
        ind[id++] = lx; ind[id++] = rx;
    }
}

```

```

        scan[sid++] = make_tuple(ly, 1, lx, rx);
        scan[sid++] = make_tuple(ry, -1, lx, rx);
    }
    int solve(){
        sort(ind.begin(), ind.end());
        ind.resize(unique(ind.begin(), ind.end()) - ind
            .begin()); // 離散化
        sort(scan, scan + sid);
        int area = 0, pre = get<0>(scan[0]);
        for(int i = 0; i < sid; i++){
            auto [x, v, l, r] = scan[i];
            area += tree[l].second * (x-pre); //
            upd(1, 0, ind.size()-1, lower_bound(ind.
                begin(), ind.end(), l)-ind.begin(),
                lower_bound(ind.begin(), ind.end(), r)-
                    ind.begin()-1, v);
            pre = x;
        }
        return area;
    }
} rect;

```

10 Math

10.1 basic

```

[a,b] 有包含
(a,b) 沒包含
ll mul(ll x,ll y,ll mod){
    ll ret=x*y-(ll)((long double)x/mod*y)*mod;
    // LL ret=x*y-(LL)((long double)x*y/mod+0.5)*mod;
    return ret<0?ret+mod:ret;
}
ll add(int x,int y,int mod){
    return (x+y)%mod;
}

```

10.2 快速幂

```

int ksm(int x, int y) {
    int ans = 1;
    while (y) {
        if (y & 1)
            ans = ans * x;
        x = x * x;
        y >>= 1;
    }
    return ans;
}

// 矩陣乘法
vector<vector<int>> mul(const vector<vector<int>> &A,
    const vector<vector<int>> &B) {
    if (A.empty() || B.empty() || A[0].empty() || B[0].
        empty()) { return {}; }
    int m = A.size();
    int k = A[0].size();
    int n = B[0].size();
    vector<vector<int>> C(m, vector<int>(n, 0));
    for (int i = 0; i < m; i++) {
        for (int p = 0; p < k; p++) if (A[i][p]) {
            for (int j = 0; j < n; j++) {
                C[i][j] = (C[i][j] + 1LL * A[i][p] * B[p][j])%MOD;
            }
        }
    }
    return C;
}

// 矩陣快速幂
vector<vector<int>> ksm(vector<vector<int>> base, int
    exp) {
    int n = base.size();
    vector<vector<int>> res(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) res[i][i] = 1;

    while (exp > 0) {
        if (exp & 1) res = mul(res, base);
        base = mul(base, base);
        exp >>= 1;
    }
    return res;
}

```

10.3 大數運算

```
//加法
string add(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        carry = 0;
    while (i >= 0 || j >= 0 || carry) {
        int sum = carry + (i >= 0 ? num1[i--] - '0' : 0) + (j >= 0 ? num2[j--] - '0' : 0);
        carry = sum / 10;
        res += to_string(sum % 10);
    }
    reverse(res.begin(), res.end());
    return res;
}

//減法
string sub(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        borrow = 0;
    while (i >= 0) {
        int sub = (num1[i--] - '0') - (j >= 0 ? num2[j--] - '0' : 0) - borrow;
        if (sub < 0) {
            sub += 10;
            borrow = 1;
        } else {
            borrow = 0;
        }
        res += to_string(sub);
    }
    while (res.length() > 1 && res.back() == '0') res.
        pop_back(); // 去掉前導0
    reverse(res.begin(), res.end());
    return res;
}

//乘法
string mul(string num1, string num2) {
    int n = num1.size(), m = num2.size();
    vector<int> res(n + m, 0);

    for (int i = n - 1; i >= 0; i--) {
        for (int j = m - 1; j >= 0; j--) {
            int p = (num1[i] - '0') * (num2[j] - '0');
            int sum = p + res[i + j + 1];

            res[i + j + 1] = sum % 10;
            res[i + j] += sum / 10;
        }
    }
    string ans = "";
    for (int x : res) {
        if (!(ans.empty() && x == 0)) // 跳過前導0
            ans += (char)(x + '0');
    }
    return ans.empty() ? "0" : ans;
}

//除法
bool smaller(string a, string b) {
    if (a.size() != b.size()) return a.size() < b.size();
    return a < b;
}

string div(string num1, string num2) {
    string cur = "", ans = "";
    for (char c : num1) {
        cur += c;
        // 去掉前導0
        while (cur.size() > 1 && cur[0] == '0') cur.
            erase(cur.begin());

        int x = 0;
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
            x++;
        }
        ans += (char)(x + '0');
    }
}
```

```
// 去掉前導0
while (ans.size() > 1 && ans[0] == '0') ans.erase(
    ans.begin());
return ans;
}

string mod(string num1, string num2) { // 餘數
    string cur = "";

    for (char c : num1) {
        cur += c;
        while (cur.size() > 1 && cur[0] == '0') cur.
            erase(cur.begin());
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
        }
    }
    return cur;
}
```

10.4 Prime

```
/* 12721, 13331, 14341, 75577, 123457, 222557, 556679
 * 999983, 1097774749, 1076767633, 100102021, 999997771
 * 1001010013, 1000512343, 987654361, 999991231
 * 999888733, 98789101, 987777733, 999991921, 1010101333
 * 1010102101, 1000000000039, 100000000000037
 * 2305843009213693951, 4611686018427387847
 * 9223372036854775783, 18446744073709551557 */
int mu[ N ], p_tbl[ N ];
vector<int> primes;
void sieve() {
    mu[ 1 ] = p_tbl[ 1 ] = 1;
    for( int i = 2 ; i < N ; i ++ ){
        if( !p_tbl[ i ] ){
            p_tbl[ i ] = i;
            primes.push_back( i );
            mu[ i ] = -1;
        }
        for( int p : primes ){
            int x = i * p;
            if( x >= M ) break;
            p_tbl[ x ] = p;
            mu[ x ] = -mu[ i ];
            if( i % p == 0 ){
                mu[ x ] = 0;
                break;
            }
        }
    }
}

vector<int> factor( int x ){
    vector<int> fac{ 1 };
    while( x > 1 ){
        int fn = SZ(fac), p = p_tbl[ x ], pos = 0;
        while( x % p == 0 ){
            x /= p;
            for( int i = 0 ; i < fn ; i ++ )
                fac.PB( fac[ pos ++ ] * p );
        }
    }
    return fac;
}
```

10.5 ExGCD

```
//ax+by=gcd(a,b)
PII gcd(int a, int b){
    if(b == 0) return {1, 0};
    PII q = gcd(b, a % b);
    return {q.second, q.first - q.second * (a / b)};
}
```

10.6 Phi

```
//歐拉函數 phi(n) 是指<=n中與n互質的數的個數
int _phi(int n) { // O(sqrtN)
    int res = n, a = n;
    for (int i = 2; i * i <= a; i++) {
        if (a % i == 0) {
            res = res / i * (i - 1);
            while (a % i == 0) a /= i;
        }
    }
    if (a > 1) res = res / a * (a - 1);
    return res;
}
```

```

int phi[MXN]; // 建表 最大 1e7
void phi_table(int n) {
    phi[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (phi[i]) continue;
        for (int j = i; j <= n; j += i) {
            if (phi[j] == 0) phi[j] = j;
            phi[j] = phi[j] / i * (i - 1);
        }
    }
}

int prime[350000]; ll phi[5000005];
bool v[5000005];
#define N 5000000
void quick_euler() {
    int cnt = 0;
    for(int i = 2; i <= N; ++i) {
        if(!v[i]) prime[++cnt] = i, phi[i] = i - 1; //
        若 i 是質數，ϕ(i) = i - 1
        for(int j = 1; i * prime[j] <= N && j <= cnt; ++j) {
            v[i * prime[j]] = 1; // 不是質數就篩掉
            if(i % prime[j] == 0){
                phi[i * prime[j]] = phi[i] * prime[j];
                break;
            }
            else phi[i * prime[j]] = phi[i] * (prime[j] - 1); // 積性函數性質用在這裡
        }
    }
}

```

10.7 Euler/歐拉函數

```

int prime[350000]; ULL phi[5000005];
bool v[5000005];
#define N 5000000
void quick_euler() {
    int cnt = 0;
    for(int i = 2; i <= N; ++i) {
        if(!v[i]) prime[++cnt] = i, phi[i] = i - 1; //
        若 i 是質數，ϕ(i) = i - 1
        for(int j = 1; i * prime[j] <= N && j <= cnt; ++j) {
            v[i * prime[j]] = 1; // 不是質數就篩掉
            if(i % prime[j] == 0){
                phi[i * prime[j]] = phi[i] * prime[j];
                break;
            }
            else phi[i * prime[j]] = phi[i] * (prime[j] - 1); // 積性函數性質用在這裡
        }
    }
}

```

10.8 miller rabin/判斷質數

```

// n < 4,759,123,141      3 : 2, 7, 61
// n < 1,122,004,669,633  4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383  6 : pimes <= 13
// n < 2^64               7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
// Make sure testing integer is in range [2, n-2] if
// you want to use magic.
ll magic[] = {};
bool witness(ll a, ll n, ll u, int t) {
    if(!a) return 0;
    ll x = mypow(a, u, n); // ksm(x, y, mod)
    for(int i = 0; i < t; i++) {
        ll nx = mul(x, x, n);
        if(nx == 1 && x != 1 && x != n-1) return 1;
        x = nx;
    }
    return x != 1;
}

bool miller_rabin(ll n) {
    int s = (7);
    // iterate s times of witness on n
    if(n < 2) return 0;

```

```

if(!(n&1)) return n == 2;
ll u = n-1; int t = 0;
// n-1 = u*2^t
while(!(u&1)) u >>= 1, t++;
while(s--){
    ll a = magic[s] % n;
    if(witness(a, n, u, t)) return 0;
}
return 1;
}

```

10.9 Pollard Rho/質因數分解

```

// 回傳 n 的其中一個因數
// does not work when n is prime 0(n^(1/4))
#define LL __int128
LL f(LL x, LL c, LL mod) { return add(mul(x, x, mod), c, mod); }
LL pollard_rho(LL n) {
    LL c = 1, x = 0, y = 0, p = 2, q, t = 0;
    while (t++ % 128 || gcd(p, n) == 1) {
        if (x == y) c++, y = f(x = 2, c, n);
        if (q = mul(p, abs(x-y), n)) p = q;
        x = f(x, c, n); y = f(f(y, c, n), c, n);
    }
    return gcd(p, n);
}

/*function<void(int)> fact=&](int x){
    if(x==1) return;
    if(miller_rabin(x)){
        f.push_back(x);return;}
    int tmp=pollard_rho(x);
    fact(tmp);fact(x/tmp);};
*/

```

10.10 FFT/傅立葉轉換 (多項式相乘)

```

const int MAXN = 262144 << 1;
// (must be 2^k)
// before any usage, run pre_fft() first
typedef long double ld;
typedef complex<ld> cplx; // real(), imag()
const ld PI = acosl(-1);
const cplx I(0, 1);
cplx omega[MAXN+1];
void pre_fft() {
    for(int i = 0; i <= MAXN; i++)
        omega[i] = exp(i * 2 * PI / MAXN * I);
}

// n must be 2^k
void fft(int n, cplx a[], bool inv=false) {
    int basic = MAXN / n;
    int theta = basic;
    for (int m = n; m >= 2; m >>= 1) {
        int mh = m >> 1;
        for (int i = 0; i < mh; i++) {
            cplx w = omega[inv ? MAXN - (i * theta % MAXN) : i * theta % MAXN];
            for (int j = i; j < n; j += m) {
                int k = j + mh;
                cplx x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
            theta = (theta * 2) % MAXN;
        }
    }
    int i = 0;
    for (int j = 1; j < n - 1; j++) {
        for (int k = n >> 1; k > (i ^= k); k >>= 1);
        if (j < i) swap(a[i], a[j]);
    }
    if(inv) for (i = 0; i < n; i++) a[i] /= n;
}

cplx arr[MAXN+1];
inline void mul(int _n, ll a[], int _m, ll b[], ll ans[]) {
    int n = 1, sum = _n + _m - 1;
    while(n < sum)
        n <<= 1;
    for(int i = 0; i < n; i++) {
        double x = (i < _n ? a[i] : 0), y = (i < _m ? b[i] : 0);
        arr[i] = complex<double>(x + y, x - y);
    }
}

```

```
fft(n,arr);
for(int i=0;i<n;i++)
    arr[i]=arr[i]*arr[i];
fft(n,arr,true);
for(int i=0;i<sum;i++)
    ans[i]=(ll)(arr[i].real()/4+0.5);
}
```

10.11 josephus/約瑟夫問題

```
int josephus(int n, int m){ //n人每m次
    int ans = 0;
    for (int i=1; i<=n; ++i)
        ans = (ans + m) % i;
    return ans;
}
```

10.12 指數塔

```
ll mod(ll x, ll m) {
    return x >= m ? (x % m) + m : x;
}
ll solve(int idx, ll m) {
    if (m == 1) return 1; // 模數降為 1
    if (idx == n) return mod(a[idx], m); // 已經到達塔頂
    ll nxt = solve(idx + 1, phi[m]);
    return fpow(a[idx], nxt, m); // 快速冪計算當前層
}
```

10.13 BSGS

```
// gcd(a, n) = 1, 算 a^x == b (mod n)
ll BSGS(ll a, ll b, ll n) {
    a %= n, b %= n;
    if (b == 1) return 0; // a^0 = 1
    ll m = (ll)sqrt(n) + 1; // 保險一點
    map<ll, ll> mp;

    // 拆成 a^{i * m} == b * a^j (mod n)
    ll baby = b;
    for (ll j = 0; j <= m; j++) {
        mp[baby] = j;
        baby = baby * a % n;
    }

    ll step = fpow(a, m, n);
    ll giant = step;
    for (ll i = 1; i <= m; i++) { // 計算 (a^m)^i
        if (mp.count(giant)) {
            ll j = mp[giant], res = i * m - j;
            if (res >= 0) return res;
        }
        giant = giant * step % n;
    }

    return -1;
}
```

10.14 $x^2 \pmod p == a$

```
void calcH(LL &t, LL &h, const LL p) {
    LL tmp=p-1; for(t=0;(tmp&1)==0;tmp/=2) t++; h=tmp;
}
// solve equation x^2 mod p = a
bool solve(LL a, LL p, LL &x, LL &y) {
    if(p == 2) { x = y = 1; return true; }
    int p2 = p / 2, tmp = mypow(a, p2, p);
    if (tmp == p - 1) return false;
    if ((p + 1) % 4 == 0) {
        x=mypow(a,(p+1)/4,p); y=p-x; return true;
    } else {
        LL t, h, b, pb; calcH(t, h, p);
        if (t >= 2) {
            do {b = rand() % (p - 2) + 2;
                while (mypow(b, p / 2, p) != p - 1);
                pb = mypow(b, h, p);
            } while (1);
            int s = mypow(a, h / 2, p);
            for (int step = 2; step <= t; step++) {
                int ss = (((LL)(s * s) % p) * a) % p;
                for(int i=0;i<t-step;i++) ss=mul(ss,ss,p);
            }
        }
    }
}
```

```
if (ss + 1 == p) s = (s * pb) % p;
pb = ((LL)pb * pb) % p;
} x = ((LL)s * a) % p; y = p - x;
} return true;
}
```

10.15 Theorem

- Lucas's Theorem :
For $n, m \in \mathbb{Z}^*$ and prime P , $C(m, n) \pmod P = \prod C(m_i, n_i)$ where m_i is the i -th digit of m in base P .
- Stirling approximation :
$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$
- Stirling Numbers(permutation $|P| = n$ with k cycles):
$$S(n, k) = \text{coefficient of } x^k \text{ in } \prod_{i=0}^{n-1} (x+i)$$
- Stirling Numbers(Partition n elements into k non-empty set):
$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$
- Pick's Theorem : $A = i + b/2 - 1$
 A : Area; i : grid number in the inner; b : grid number on the side
- Catalan number : $C_n = \binom{2n}{n} / (n+1)$
$$C_n^{n+m} - C_{n+1}^{n+m} = (m+n)! \frac{n-m+1}{n+1} \quad \text{for } n \geq m$$

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = 2 \binom{2n+1}{n+2} C_n$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = \sum_{i=0}^n C_i C_{n-i} \quad \text{for } n \geq 0$$
- Euler Characteristic:
planar graph: $V - E + F - C = 1$
convex polyhedron: $V - E + F = 2$
 V, E, F, C : number of vertices, edges, faces(regions), and components
- Kirchhoff's theorem :
 $A_{ii} = \deg(i), A_{ij} = (i, j) \in E ? - 1 : 0$, Deleting any one row, one column, and cal the $\det(A)$
- Polya' theorem (c is number of color, m is the number of cycle size):
$$\left(\sum_{i=1}^m c^{\gcd(i, m)} \right) / m$$
- Burnside lemma:
$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$
- 錯排公式: (n 個人中, 每個人皆不再原來位置的組合數):
$$dp[0] = 1; dp[1] = 0;$$

$$dp[i] = (i-1) * (dp[i-1] + dp[i-2]);$$
- Bell 數 (有 n 個人, 把他們拆組的方法總數) :
$$B_0 = 1$$

$$B_n = \sum_{k=0}^n s(n, k) \quad (\text{second-stirling})$$

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$$
- Wilson's theorem :
$$(p-1)! \equiv -1 \pmod p$$
- Fermat's little theorem :
$$a^p \equiv a \pmod p$$
- Euler's totient function:
$$A^{B^C} \pmod p = \text{pow}(A, \text{pow}(B, C, p-1)) \pmod p$$
- 歐拉函數降冪公式:
$$A^{B^C} \pmod C = A^{B \pmod{\phi(C) + \phi(C)}} \pmod C$$
- 6 的倍數:
$$(a-1)^3 + (a+1)^3 + (-a)^3 + (-a)^3 = 6a$$
- Standard young tableau (標準楊表):
 $\lambda = (\lambda_1 \geq \dots \geq \lambda_k), \sum \lambda_i = n$ denoted by $\lambda \vdash n$
 $\lambda \vdash n$ 意思為 λ 整數拆分 n eg. $n = 10, \lambda = (6, 4)$ 此拆分可表示一種楊表形狀。
楊表: 第 1 列 λ_1 行... 第 k 列 λ_k 行的方格圖。
標準楊表: 每列從左到右遞增; 每行從上到下遞增。
Let T 為某一 Permutation 跑 RSK 後的標準楊表, 則此 Permutation 的 LDS、LIS 長度分別為 T 的列、行數。
- RSK Correspondence:
A permutation is bijective to (P, Q) 一對標準楊表
 P : Permutation 跑 RSK 算法的結果, 可為半標準楊表。
 Q : 可用來還原 Permutation (像排列矩陣)。
- Hook length formula (形狀為 λ 的標準楊表個數):
$$f^\lambda = \frac{n!}{\prod h_\lambda(i, j)}$$

$$h_\lambda(i, j) = \text{number of pair } (x, y) \text{ where } (x = i \vee y = j) \wedge (x, y) \geq (i, j)$$

且 (x, y) 落在形狀為 λ 的表上。
Recursion:
(i) $f^{(0, \dots, 0)} = 1$
(ii) $f^{(\lambda_1, \dots, \lambda_m)} = \sum_{k=1}^m f^{(\lambda_1, \dots, \lambda_{k-1}, \lambda_k-1, \lambda_{k+1}, \dots, \lambda_m)}$

- 一個環上相鄰顏色不同的染色數量：
 $f(n, k) = (k - 1)^n + (-1)^n (k - 1)$
n 是點的數量，k 是顏色的數量





